BOSTON HOUSING

23001851 Wei-chung Cheng

23003115 José Carlos Gordillo Alarcón

23001127 Samuel Ernesto Caal Ramirez

1. INTRODUCTION

Accurate housing price prediction is a fundamental challenge in real estate analytics and machine learning. Property values are influenced by multiple factors, including neighborhood characteristics, environmental conditions, accessibility, and socioeconomic indicators. The Boston Housing dataset serves as a benchmark dataset for regression tasks, providing valuable information about residential areas in Boston. This project evaluates and compares different machine learning models, including MLP, SVR, Random Forest, and XGBoost, to predict median house values. Model performance was assessed using standard regression metrics to determine the most effective approach for this prediction problem.

2. DATASET DESCRIPTION

The Boston Housing dataset is derived from the information collected by the U.S. Census Service concerning housing in the area of Boston in 1970. It consists of a total of 506 samples, each containing 12 numeric features that explains one numeric target variable. The dataset has no missing values.

Feature	Category	Description	Unit / Type	Mean	Std.
MEDV	Target	Median home value	In $1,000	22.53	9.197
CRIM	Neighborhood	Per capita crime rate	Rate	3.614	8.602
ZN	Neighborhood	Large residential zoning (>25ksq.ft.)	Percentage	11.36	23.32
INDUS	Neighborhood	Non-retail industrial acres	Percentage	11.14	6.860
CHAS	Neighborhood	Bounds the Charles River	Binary (0 / 1)	0.069	0.254
AGE	Neighborhood	Homes built before 1940	Percentage	68.57	28.15
LSTAT	Neighborhood	Lower socioeconomic status population	Percentage	12.65	7.141
RM	Property	Average rooms per dwelling	Count	6.285	0.703
NOX	Environment	Nitric oxides concentration	Parts per 10M	0.554	0.116
DIS	Location	Distance to 5 Boston employment centers	Weighted distance	3.795	2.106
RAD	Location	Accessibility to radial highways	Index (1-24)	9.549	8.707
TAX	Financial	Property tax rate	Per $10,000	408.2	168.5
PTRATIO	Education	Pupil-teacher ratio	Ratio	18.46	2.165

3. METHODOLOGY

We designed a predictive pipeline to model Boston housing prices (MEDV) by benchmarking three classical machine learning algorithms against the simplest deep learning architecture, a multilayer perceptron.

Data preprocessing & validation split

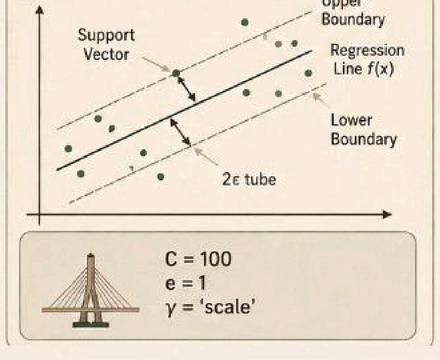
- Data normalization. The columns CRIM, ZN, AGE, DIS, RAD, TAX and LSTAT were standard scaled (mean of 0 and standard deviation of 1) with Z normalization to keep a similar magnitude between the features.
- Validation strategy. To guarantee fair benchmarking, all models were trained and tested on the same train-test ratio, as well as the same random state for the splitting of the dataset. Furthermore, a strict 5-fold Cross Validation strategy was enforced across all models, ensuring identical evaluation subsets.

Data preprocessing → K-Fold CV Hyperparameter tuning → Multi-Metric Training

K-Fold Cross Validation & Hyperparameter tuning

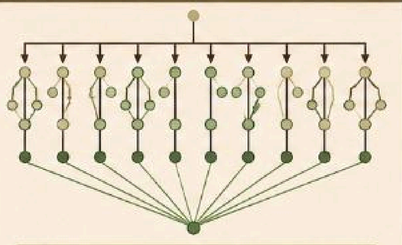
K-fold cross validation (CV) was used to evaluate the robustness of the models and estimate their predictive performance. For the classical ML algorithms, it was used to determine the optimal model and the corresponding hyperparameters that bring the best results. For the neural network, CV was used to find the hyperparameters that offer a good tradeoff between generalization and performance. Afterwards, the selected hyperparameters were used to train with the complete training set for the final model.

1. SVR ARCHITECTURE



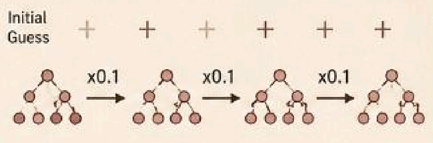
C = 100
e = 1
γ = 'scale'

2. RANDOM FOREST REGRESSOR



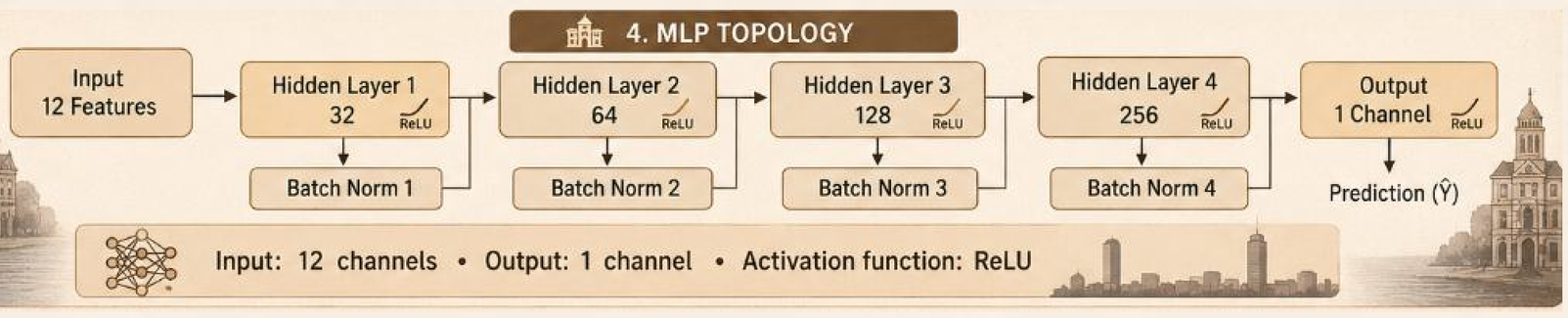
n_estimators = 100
max_depth = None
max_features = None
min_samples_split = 2

3. XGBOOST (GRADIENT BOOSTING)



n_estimators = 200
learning_rate = 0.1
max_depth = 3
subsample = 0.9
colsample_bytree = 0.8

4. MLP TOPOLOGY

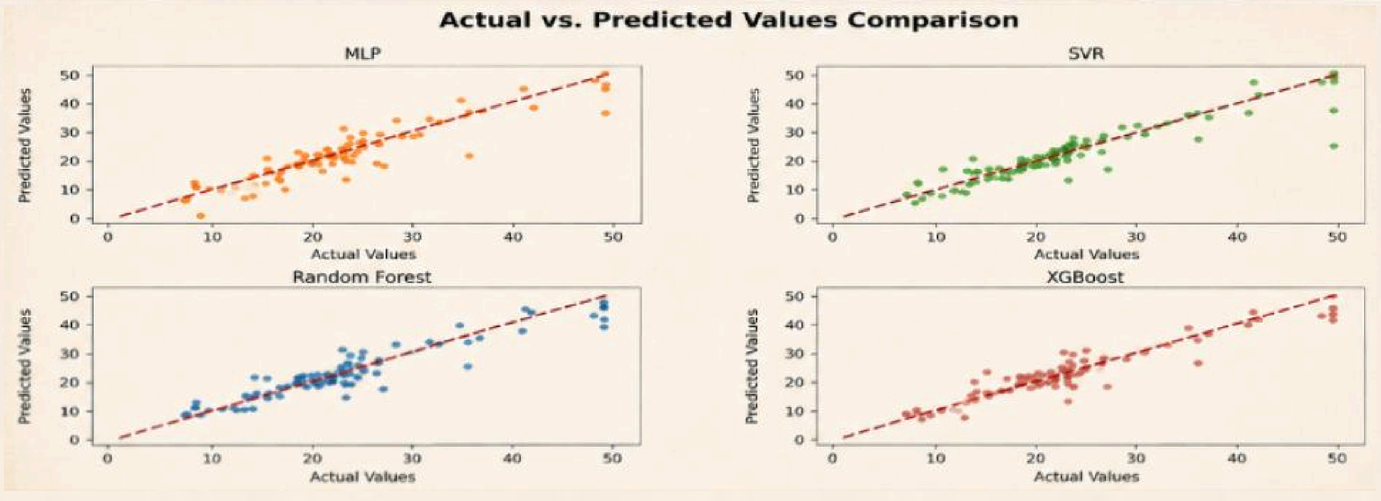


4. RESULTS

The models were evaluated across three regression performance metrics. The tree ensemble methods outperformed both SVR and the deep learning approach, with XGBoost achieving the highest overall prediction accuracy.

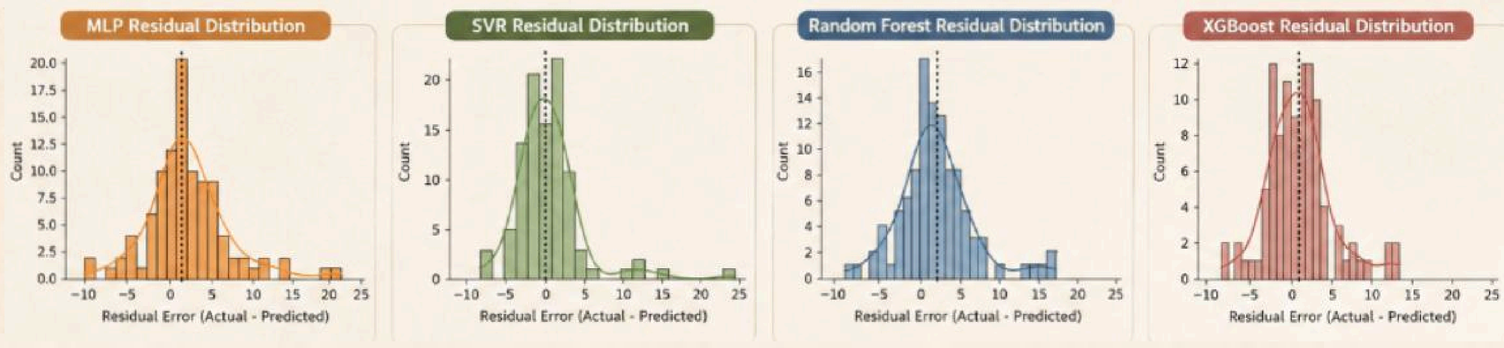
The Actual vs. Predicted plot shows clusters tightly aligned with the identity line, proving a high prediction accuracy along the dataset.

Random Forest and XGBoost show a tight, symmetric Gaussian bell centered at 0, meaning they have consistent error variance across all values. Meanwhile, MLP and SVR show wider, more distributed tails, meaning a few severe outliers in the predictions.



MODEL PERFORMANCE COMPARISON

Model	MAE	RMSE	R2
MLP	2.568	3.758	0.8459
SVR	2.332	3.837	0.8488
Random Forest	2.371	3.338	0.8855
XGBoost	2.312	3.184	0.8958



4. RESULTS

- Tree ensemble models proved to be the most robust architectures for the current problem. They successfully captured the non-linear relationships between the features without requiring extensive hyperparameter tuning.
- The models were fundamentally constrained by the dataset's price ceiling at $50k. While the tree ensemble models could handle the imposed limitation, the SVR and MLP models suffered from mispredictions because of this boundary. Having this restriction in the training data also means that these models could underperform when working with external data.
- Despite incorporating modern architecture stability method, batch normalization, the MLP struggled to outperform the simpler tree ensemble models. This confirms that for smaller tabular datasets, gradient boosting machines remain more effective and efficient against deep neural networks.

5. CONCLUSIONS AND FUTURE IMPROVEMENTS

- Address the artificial $50k truncation cap found in the dataset. Given that the models are learning the underlying patterns of the data, we want to prevent them from learning biased boundaries.
- House prices are usually found in a right-skewed distribution. Because of this, an adequate transformation to the dependent variable before training can stabilize the variance and help the models make cleaner predictions.
- Since each of the models used in this work learns a completely different pattern representation of the data, a model ensemble can be used to combine each of their individual predictions to generate a lower variance result.
- Integrate tools, such as optimization framework, orchestration tool and experiment trackers, to make an automated workflow of a reproducible pipeline, with automatic logging and plotting, ensuring perfect reproducibility.

Galileo UNIVERSIDAD

MACHINE LEARNING MODELS FOR HOUSE PRICE PREDICTION

DATA SCIENCE PROJECT

SCAN ME FOR MORE INFO

- Full Project Report
- Source Code
- Dataset
- Additional Results